Project Initialization and Planning Phase

Date	27 June 2024
Team ID	
Project Title	Rising Waters
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution aims to improve flood prediction using machine learning techniques. The system analyzes historical weather data, including rainfall, cloud visibility, and seasonal rainfall patterns, to predict the likelihood of floods. By providing early and accurate predictions, the system helps disaster management authorities take timely action, reduce risks, and improve public safety.

Project Overview	
Objective	The primary objective is to develop an intelligent flood prediction system using machine learning algorithms to provide accurate early flood warnings and support disaster preparedness.
Scope	The project predicts flood risk using historical weather data and machine learning models through a Flask web application for early warning.
Problem Statement	
Description	Traditional flood forecasting methods often fail to provide timely and accurate predictions, resulting in delayed warnings, property damage, and loss of life.
Impact	An accurate flood prediction system enables early evacuation, better disaster planning, efficient resource allocation, and reduced damage to lives and property.
Proposed Solution	
Approach	Develop a machine learning model to analyze historical weather data and predict flood risk. Deploy the best-performing model using a Flask web application.
Key Features	-Machine learning-based flood prediction -Historical weather data Analysis

	-Flask web application -Early flood risk alerts.

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i5/i7 Processor (GPU not required)
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	256 GB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, XGBoost, numpy, matplotlib
Development Environment	IDE	Jupyter Notebook, VS Code
Data		
Data	Source, size, format	Kaggle Flood Prediction Dataset, CSV